DATA ANALYTICS INTERNSHIP - InternSpark

(AICTE-Listed Internship Program)

REPORT

ON

SALES PERFORMANCE ANALYSIS

(TASK 2)

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This report analyzes the Superstore sales dataset to evaluate business performance across regions, product categories, and time. The analysis highlights key trends such as strong performance in the West region, high sales contribution from the Technology category, and noticeable seasonal peaks during the end of the year.
The study also identifies that a small number of products generate a significant portion of total revenue, while some products and regions underperform. Customer segmentation shows that the Consumer segment contributes the most to overall sales, indicating a strong opportunity for targeted marketing.
Based on these insights, several tactical improvements have been recommended, including focusing on high-performing products, adopting region-specific strategies, leveraging seasonal demand, optimizing the product portfolio, and implementing customer-centric approaches.
Overall, the findings emphasize the importance of data-driven decision-making to enhance sales performance, improve operational efficiency, and support business growth for Alfido Tech.

INTRODUCTION

This project analyzes the Superstore sales dataset to evaluate business performance across regions, product categories, and time. The goal is to identify trends, top-performing products, and seasonal patterns to support data-driven decision-making for Alfido Tech.

DATASET OVERVIEW

The dataset contains transactional sales data with the following key attributes:

- Order ID, Order Date, Ship Date
- Customer details (Customer ID, Name, Segment)
- Location (Country, City, State, Region)
- Product details (Category, Sub-Category, Product Name)
- Sales amount

This data helps in understanding customer behavior, regional performance, and product demand.

DATA PREPARATION

The dataset was cleaned and prepared using the following steps:

1. Converted **Order_Date** and **Ship_Date** into datetime format
2. Removed missing and duplicate records
3. Standardized column names
4. Created new features:
    - Year
    - Month
    - Month Name

These steps ensured accurate and structured analysis.

KEY PERFORMANCE INDICATORS (KPIs)

The following KPIs were calculated:

- **Total Revenue:** Sum of all sales values
- **Total Orders:** Number of unique orders
- **Average Order Value (AOV):** Average revenue per order

These KPIs provide a high-level overview of business performance.

SALES PERFORMANCE ANALYSIS

1. **REGIONAL PERFORMANCE** : Indicates the need for targeted strategies in underperforming regions.
   - The **West region** generated the highest sales.
   - The **Central region** showed comparatively lower performance.

2. **CATEGORY ANALYSIS** : Technology products are the primary revenue drivers.
   - **Technology** category contributed the highest sales.
   - **Furniture** and **Office Supplies** followed.

3. **SUB-CATEGORY INSIGHTS** : Suggests uneven product performance.
   - A few sub-categories dominate total sales.
   - Some sub-categories contribute very little.

4. **TOP PERFORMING PRODUCTS** :
   - Top 10 products contribute a significant share of total revenue.
   - High demand products should be prioritized in marketing and inventory.

5. **LOW PERFORMING PRODUCTS** :
   - Certain products consistently generate low sales.
   - These products may need improvement, repositioning, or removal.

TIME-BASED ANALYSIS (SEASONALITY)

- Sales show strong variation across months
- Peak sales observed during **November and December**
- Lower sales seen in early months of the year.

Indicates strong **seasonal demand patterns**, likely due to holidays and promotions.

CUSTOMER SEGMENT ANALYSIS

- The **Consumer segment** contributes the largest share of sales
- Corporate and Home Office segments follow

Consumer-focused strategies can drive further growth.

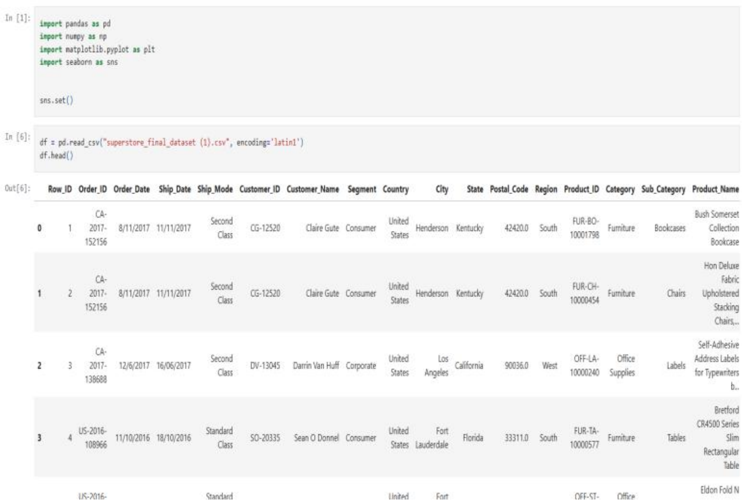
SHIPPING ANALYSIS

- Standard shipping mode accounts for most orders
- Faster shipping modes are used less frequently

Customers prefer cost-effective shipping options.

SCREENSHOTS AND VISUALIZATIONS

1.



In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


sns.set()
```

In [6]:

```
df = pd.read_csv("superstore_final_dataset (1).csv", encoding='latin1')
df.head()
```

Out[6]:

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
		US-2016-			Standard				United	Fort				OFF-ST-	Office		Eldon Fold N

2.

In [7]:
```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Row_ID          9800 non-null   int64
 1   Order_ID        9800 non-null   object
 2   Order_Date      9800 non-null   object
 3   Ship_Date       9800 non-null   object
 4   Ship_Mode       9800 non-null   object
 5   Customer_ID     9800 non-null   object
 6   Customer_Name   9800 non-null   object
 7   Segment         9800 non-null   object
 8   Country         9800 non-null   object
 9   City            9800 non-null   object
 10  State           9800 non-null   object
 11  Postal_Code     9789 non-null   float64
 12  Region          9800 non-null   object
 13  Product_ID      9800 non-null   object
 14  Category        9800 non-null   object
 15  Sub_Category    9800 non-null   object
 16  Product_Name    9800 non-null   object
 17  Sales           9800 non-null   float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
```

Out[7]:

	Row_ID	Postal_Code	Sales
count	9800.000000	9789.000000	9800.000000
mean	4900.500000	55273.322403	230.769059
std	2829.160653	32041.223413	626.651875
min	1.000000	1040.000000	0.444000

3.

In [9]:
```
df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True)
df['Ship_Date'] = pd.to_datetime(df['Ship_Date'], dayfirst=True)
```

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
Row_ID            0
Order_ID          0
Order_Date        0
Ship_Date         0
Ship_Mode         0
Customer_ID       0
Customer_Name     0
Segment           0
Country           0
City              0
State             0
Postal_Code      11
Region            0
Product_ID        0
Category          0
Sub_Category      0
Product_Name      0
Sales             0
dtype: int64
```

In [11]:
```
df = df.dropna()
```

In [12]:
```
df = df.drop_duplicates()
```

In [13]:
```
df['Year'] = df['Order_Date'].dt.year
df['Month'] = df['Order_Date'].dt.month
df['Month_Name'] = df['Order_Date'].dt.strftime('%B')
```

4.

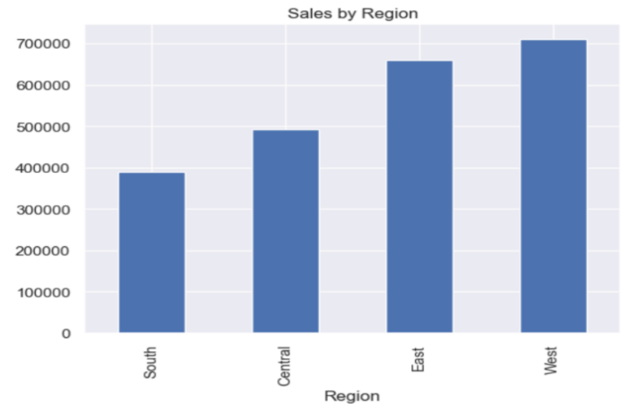
In [14]:
```
month_order = ['January','February','March','April','May','June',
               'July','August','September','October','November','December']

df['Month_Name'] = pd.Categorical(df['Month_Name'], categories=month_order, ordered=True)
```

In [15]:
```
total_revenue = df['Sales'].sum()
avg_order_value = df.groupby('Order_ID')['Sales'].sum().mean()
total_orders = df['Order_ID'].nunique()

print("Total Revenue:", total_revenue)
print("Average Order Value:", avg_order_value)
print("Total Orders:", total_orders)
```

```
Total Revenue: 2252607.4127
Average Order Value: 458.2195713384866
Total Orders: 4916
```

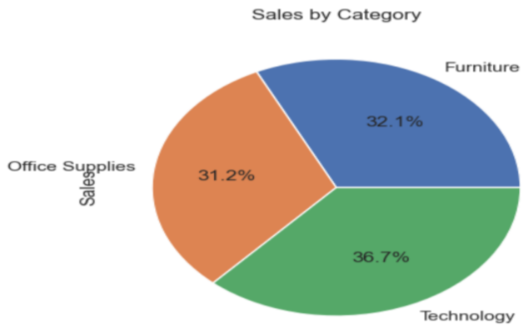
5.

```
In [16]: region_sales = df.groupby('Region')['Sales'].sum().sort_values()

         region_sales.plot(kind='bar', title='Sales by Region')
         plt.show()
```



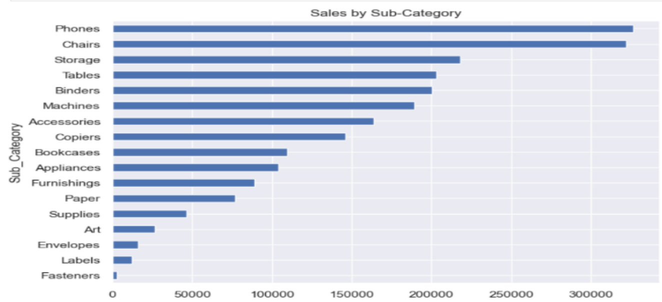
6.

```
In [17]: category_sales = df.groupby('Category')['Sales'].sum()

         category_sales.plot(kind='pie', autopct='%1.1f%%')
         plt.title("Sales by Category")
         plt.show()
```



7.

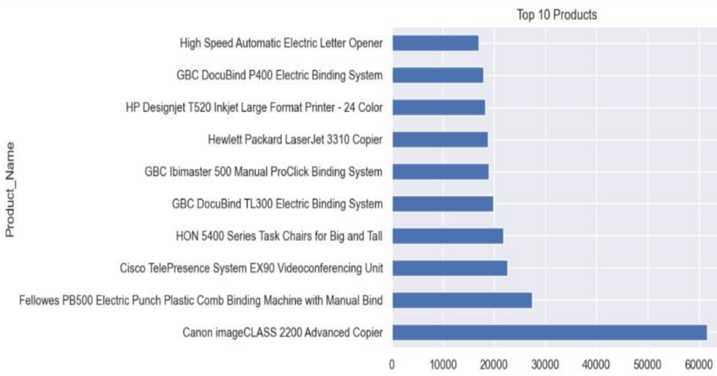
```
In [18]: subcat_sales = df.groupby('Sub_Category')['Sales'].sum().sort_values()

         subcat_sales.plot(kind='barh', figsize=(8,6))
         plt.title("Sales by Sub-Category")
         plt.show()
```



8.

In [19]:

```python
top_products = df.groupby('Product_Name')['Sales'].sum().sort_values(ascending=False).head(10)

top_products.plot(kind='barh')
plt.title("Top 10 Products")
plt.show()
```



9.

In [22]:

```python
monthly_sales = df.groupby('Month_Name')['Sales'].sum()

monthly_sales.plot(kind='line', marker='o')
plt.title("Monthly Sales Trend")
plt.show()
```

```
C:\Users\shiva\AppData\Local\Temp\ipykernel_22760\4024411788.py:1: FutureWarning: The
observed=False to retain current behavior or observed=True to adopt the future default
  monthly_sales = df.groupby('Month_Name')['Sales'].sum()
```

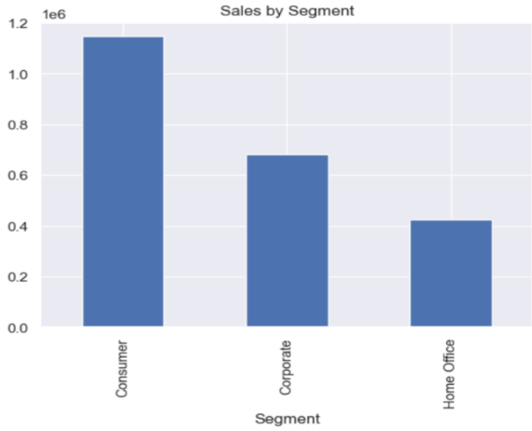
10.

In [23]:

```python
yearly_sales = df.groupby('Year')['Sales'].sum()

yearly_sales.plot(kind='bar')
plt.title("Yearly Sales")
plt.show()
```

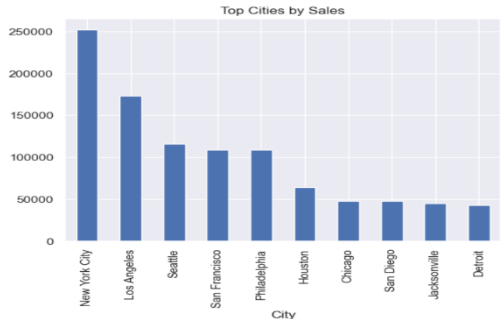
11.

```
In [24]: segment_sales = df.groupby('Segment')['Sales'].sum()

         segment_sales.plot(kind='bar')
         plt.title("Sales by Segment")
         plt.show()
```



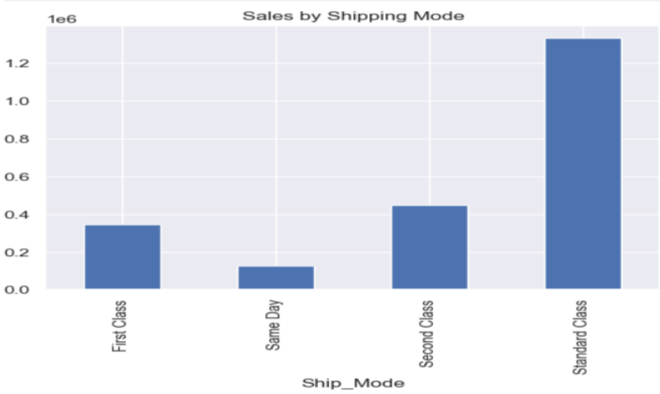
12.

```
top_cities = df.groupby('City')['Sales'].sum().sort_values(ascending=False).head(10)

top_cities.plot(kind='bar')
plt.title("Top Cities by Sales")
plt.show()
```



13.

```
ship_mode_sales = df.groupby('Ship_Mode')['Sales'].sum()

ship_mode_sales.plot(kind='bar')
plt.title("Sales by Shipping Mode")
plt.show()
```



KEY INSIGHTS

- West region is the strongest market.
- Technology is the top-performing category.
- Sales are highly seasonal (Nov–Dec peak).
- A small number of products drive most revenue.
- Some products and regions underperform significantly.

TACTICAL IMPROVEMENTS

1. **Smart Product Promotion Strategy** : Alfido Tech should focus its marketing efforts on top-performing products instead of promoting all items equally. By highlighting bestsellers and creating product bundles that combine high-demand and low-demand items, the company can increase visibility and drive higher sales conversions.
2. **Region-Specific Sales Strategy** : Sales performance varies across regions, so Alfido Tech should adopt region-specific strategies. This can include localized advertising campaigns, customized discounts, and better logistics planning to improve performance in underperforming regions.
3. **Seasonal Sales Planning** : Sales trends indicate strong seasonality, especially during peak months like November and December. Alfido Tech should prepare in advance by increasing inventory, launching festive promotions, and aligning marketing campaigns with high-demand periods to maximize revenue.
4. **Product Portfolio Optimization** : Some products consistently perform poorly and contribute little to overall sales. Alfido Tech should evaluate these products and either improve their pricing and quality or remove them from the portfolio to focus on more profitable items.
5. **Customer-Centric Approach** : Focusing on high-value customer segments can significantly improve sales. Alfido Tech can implement personalized offers, loyalty programs, and targeted recommendations based on customer purchase behavior to increase repeat purchases and customer satisfaction.

CODE LINK :

https://github.com/shivangikumari1409/InternSpark-Internship/blob/3f47b6a8c7cd51dff222357b98c9f5e7179a9168/SalesPerformaceAnalysis/Sales_Performace_Analysis.py